

Gradient-Product-Balanced Initialization

A Per-Layer Variance Schedule for Row-Centered ReLU Networks

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1 Goal and Motivation

We seek an initialization scheme for deep ReLU networks that simultaneously:

1. **Preserves geometry** — class separability should survive deep projections.
2. **Maintains gradient stability** — gradient magnitudes should be approximately uniform across layers.

From BP4, the gradient of the loss $\mathcal{C}$ at layer ℓ is:

$$\frac{\partial \mathcal{C}}{\partial W^{(\ell)}} = \frac{1}{n} (\Delta^{(\ell)})^\top A^{(\ell-1)}, \quad (1)$$

where $A^{(\ell-1)} \in \mathbb{R}^{n \times d_{\ell-1}}$ is the activation matrix (input to layer ℓ) and $\Delta^{(\ell)} \in \mathbb{R}^{n \times d_\ell}$ is the error signal matrix. Taking Frobenius norms:

$$\left\| \frac{\partial \mathcal{C}}{\partial W^{(\ell)}} \right\|_F \propto \text{rms}(A^{(\ell-1)}) \cdot \text{rms}(\Delta^{(\ell)}). \quad (2)$$

This product depends on two per-layer quantities:

- **Forward gain:** $g_{\text{fwd}}^{(\ell)} = \text{rms}(A^{(\ell)}) / \text{rms}(A^{(\ell-1)})$ — how the signal changes layer to layer.
- **Backward gain:** $g_{\text{bwd}}^{(\ell)} = \text{rms}(\Delta^{(\ell)}) / \text{rms}(\Delta^{(\ell+1)})$ — how the error signal changes backward.

We define the **gradient ratio** as $\max_\ell \|\nabla_{W^{(\ell)}} \mathcal{C}\|_F / \min_\ell \|\nabla_{W^{(\ell)}} \mathcal{C}\|_F$. A ratio of 1 means perfectly uniform gradients.

2 Forward and Backward Gains for Row-Centered Weights

2.1 Setup and notation

Consider a layer with row-centered weight matrix $W \in \mathbb{R}^{d \times d}$, meaning $\sum_j W_{ij} = 0$ for every row i . The per-element variance is $\text{Var}(W_{ij}) = v/d$. We define a convenient **variance factor**:

$$s = \sqrt{\frac{v}{2}}. \quad (3)$$

When $s = 1$, we recover standard He variance ($v = 2/d \cdot d = 2$, i.e., $\text{Var}(W_{ij}) = 2/d$).

2.2 Forward gain: why row centering loses signal

The key question: how much output energy $\mathbb{E}[(a^{(\ell)})^2]$ does a layer produce, given input energy $\mathbb{E}[(a^{(\ell-1)})^2]$?

Without row centering (standard He), the answer is well known. Each neuron computes $z_i = \sum_j W_{ij} a_j$, and the variance of z_i is:

$$\text{Var}(z_i) = d \cdot \text{Var}(W_{ij}) \cdot \mathbb{E}[a_j^2] = v \cdot \mathbb{E}[a^2].$$

After ReLU (which zeroes out $\sim 50\%$ of values), the output energy is $\mathbb{E}[(a^{(\ell)})^2] = \frac{v}{2} \cdot \mathbb{E}[(a^{(\ell-1)})^2]$. He sets $v = 2$, giving $\mathbb{E}[(a^{(\ell)})^2] = \mathbb{E}[(a^{(\ell-1)})^2]$: forward gain = 1.

With row centering, the neuron output changes. Since $\sum_j W_{ij} = 0$, we can write:

$$z_i = \sum_j W_{ij} a_j = \sum_j W_{ij} (a_j - \bar{a}), \quad \bar{a} = \frac{1}{d} \sum_k a_k.$$

The constant part $\bar{a} \cdot \sum_j W_{ij}$ vanishes. The neuron only “sees” the deviations $a_j - \bar{a}$, not the mean $\bar{a}$.

Now, post-ReLU activations $a = \text{ReLU}(z)$ with $z \sim \mathcal{N}(0, \sigma^2)$ have a positive mean:

$$\mathbb{E}[a] = \frac{\sigma}{\sqrt{2\pi}} > 0, \quad \mathbb{E}[a^2] = \frac{\sigma^2}{2}, \quad \text{Var}(a) = \mathbb{E}[a^2] - (\mathbb{E}[a])^2 = \frac{\sigma^2(\pi - 1)}{2\pi}.$$

The fraction of energy in the deviations (what row-centered weights can access) is:

$$\frac{\text{Var}(a)}{\mathbb{E}[a^2]} = \frac{\pi - 1}{\pi} \approx 0.682. \quad (4)$$

This is a universal property of ReLU — independent of σ or d .

Since row-centered weights see $\text{Var}(a)$ instead of $\mathbb{E}[a^2]$, the output energy becomes:

$$\mathbb{E}[(a^{(\ell)})^2] = \frac{v}{2} \cdot \frac{\pi - 1}{\pi} \cdot \mathbb{E}[(a^{(\ell-1)})^2] = s^2 \cdot \frac{\pi - 1}{\pi} \cdot \mathbb{E}[(a^{(\ell-1)})^2].$$

Taking square roots for the forward gain:

$$\boxed{g_{\text{fwd}} = r \cdot s, \quad r = \sqrt{\frac{\pi - 1}{\pi}} \approx 0.826.} \quad (5)$$

The factor r is structural: it comes from row centering being blind to the mean of post-ReLU activations. No variance adjustment can remove it.

2.3 Backward gain: why the backward pass is unaffected

The backward pass computes $\delta^{(\ell)} = (W^{(\ell+1)})^\top \delta^{(\ell+1)} \odot \sigma'(z^{(\ell)})$, where $\sigma'(z^{(\ell)}) = \mathbf{1}[z^{(\ell)} > 0]$ is the ReLU derivative (binary mask).

Row centering of W means $\sum_j W_{ij} = 0$ (rows sum to zero). The transpose $W^\top$ therefore has $\sum_i W_{ij} = 0$ (columns sum to zero). By the same logic as the forward pass, $W^\top$ is blind to the mean of its input $\delta^{(\ell+1)}$.

However, unlike post-ReLU activations, the error signal δ contains both positive and negative values, so its mean is near zero: $\text{Var}(\delta)/\mathbb{E}[\delta^2] \approx 1$. There is almost no “DC component” to lose.

Therefore the backward gain depends only on the weight scale:

$$\boxed{g_{\text{bwd}} = s.} \quad (6)$$

2.4 The coupling

Dividing (5) by (6):

$$\frac{g_{\text{fwd}}}{g_{\text{bwd}}} = r \approx 0.826, \quad \text{always, regardless of } s. \quad (7)$$

No single scalar variance can make both gains equal to 1 simultaneously.

3 The Product-Balanced Initializer

3.1 Derivation

Since we cannot achieve $g_{\text{fwd}} = g_{\text{bwd}} = 1$, we seek the next best thing: $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$.

Theorem 3.1 (Product-balanced variance). *The unique s^* satisfying $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$ is*

$$s^* = \left(\frac{\pi}{\pi - 1} \right)^{1/4} \approx 1.1006, \quad \text{Var}(W_{ij}) = \frac{2\sqrt{\pi/(\pi - 1)}}{d} \approx \frac{2.422}{d}. \quad (8)$$

Proof. $g_{\text{fwd}} \cdot g_{\text{bwd}} = rs \cdot s = rs^2 = 1 \implies s^2 = 1/r = \sqrt{\pi/(\pi - 1)} \implies s^* = (\pi/(\pi - 1))^{1/4}$. $\square$

Resulting gains: $g_{\text{fwd}}^* = ((\pi - 1)/\pi)^{1/4} \approx 0.909$, $g_{\text{bwd}}^* = (\pi/(\pi - 1))^{1/4} \approx 1.101$, $g_{\text{fwd}}^* \cdot g_{\text{bwd}}^* = 1$ (exact).

3.2 Initialization procedure

Definition 3.2 (Product-balanced row-centered initialization). Given a layer with fan-in d :

1. Set $\sigma = \sqrt{2\sqrt{\pi/(\pi-1)}} / d$.
2. Sample $W_{ij} \sim \mathcal{N}(0, \sigma^2)$.
3. Row-center: $W_{ij} \leftarrow W_{ij} - \frac{1}{d} \sum_k W_{ik}$.
4. Rescale each row: $\mathbf{w}_i \leftarrow \mathbf{w}_i \cdot \sigma / \text{std}(\mathbf{w}_i)$.
5. Set bias to zero.

4 Per-Layer Variance Scaling: The η -Schedule

Uniform variance cannot produce uniform gradients (as we show in Section 7). We derive a per-layer schedule from BP4.

4.1 Derivation

Step 1 (Forward chain). $\text{rms}(A^{(\ell-1)}) = \text{rms}(A^{(0)}) \cdot \prod_{k=1}^{\ell-1} (r \cdot s_k) = \text{rms}(A^{(0)}) \cdot r^{\ell-1} \cdot \prod_{k=1}^{\ell-1} s_k$.

Step 2 (Backward chain). The backward gain at position k is $g_{\text{bwd}}^{(k)} \approx s_{k+1}$. Telescoping: $\text{rms}(\Delta^{(\ell)}) = \text{rms}(\Delta^{(L)}) \cdot \prod_{k=\ell+1}^L s_k$.

Step 3 (Gradient at layer ℓ). Substituting into (2):

$$\left\| \frac{\partial \mathcal{C}}{\partial W^{(\ell)}} \right\| \propto r^{\ell-1} \cdot \frac{\prod_{k=1}^L s_k}{s_\ell}. \quad (9)$$

Since $\prod_{k=1}^L s_k$ is constant across ℓ :

$$\left\| \frac{\partial \mathcal{C}}{\partial W^{(\ell)}} \right\| \propto \frac{r^{\ell-1}}{s_\ell}. \quad (10)$$

Step 4 (Uniform gradients). $r^{\ell-1}/s_\ell = \text{const}$ requires $s_\ell \propto r^{\ell-1}$.

4.2 The η -parameterization

Setting $s_\ell = C \cdot r^{\ell-1}$ directly would fully correct the gradient, but at the cost of extreme variance differences between layers (since $r < 1$, early layers get much larger variance than late layers). We introduce a *correction strength* $\eta \in [0, 1]$:

$$s_\ell = s^* \cdot r^{\eta(\ell-(L+1)/2)}, \quad (11)$$

where $s^* = (\pi/(\pi-1))^{1/4}$ (the product-balanced factor) and the exponent is centered at the middle layer so the geometric mean equals s^* .

- $\eta = 0$: uniform variance, no correction. Gradient ratio $= r^{-(L-1)}$.
- $\eta = 1$: full correction, perfect gradient uniformity, but variance ratio $= r^{-(L-1)}$.
- $0 < \eta < 1$: partial correction — a trade-off.

The parameter η is **not derived from theory** — it is a tunable knob that controls the trade-off.

Definition 4.1 (Layer-balanced product-base initialization). For layer $\ell \in \{1, \dots, L\}$ with fan-in $d_{\ell-1}$:

$$\sigma_\ell = \sqrt{\frac{2}{d_{\ell-1}}} \cdot s^* \cdot r^{\eta(\ell-(L+1)/2)}. \quad (12)$$

Then apply the same procedure as Definition 3.2, using σ_ℓ (which varies per layer) instead of a fixed σ .

5 Empirical Validation (50 Hidden Layers)

5.1 Setup

Fashion-MNIST, 2000 samples, architecture $784 \rightarrow [784]^{50} \rightarrow 1$. Four initializers compared:

1. **He** — standard Kaiming (baseline).
2. **RC fwd-balanced** — $g_{\text{fwd}} = 1$, $g_{\text{bwd}} \approx 1.21$.
3. **RC product-balanced** — $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$ (uniform).
4. **RC layer-balanced (product base, $\eta = 0.5$)** — per-layer scaling.

5.2 Gain verification

Table 1: Per-layer gain comparison, 50 hidden layers.

Initializer	Med g_{fwd}	Med g_{bwd}	Product	Grad ratio	Grad CV
He (baseline)	0.993	0.997	0.990	$7.8\times$	0.522
RC fwd-balanced	1.000	1.209	1.210	$1901\times$	1.903
RC product-balanced	0.909	1.099	0.999	$1902\times$	1.906
RC layer-bal ($\eta=0.5$)	0.956	1.049	1.003	$36\times$	1.095

The layer-balanced variant reduces the gradient ratio from $1902\times$ (uniform RC) to $36\times$ — a $53\times$ improvement. However, He’s $7.8\times$ remains significantly better.

5.3 Forward gain per layer

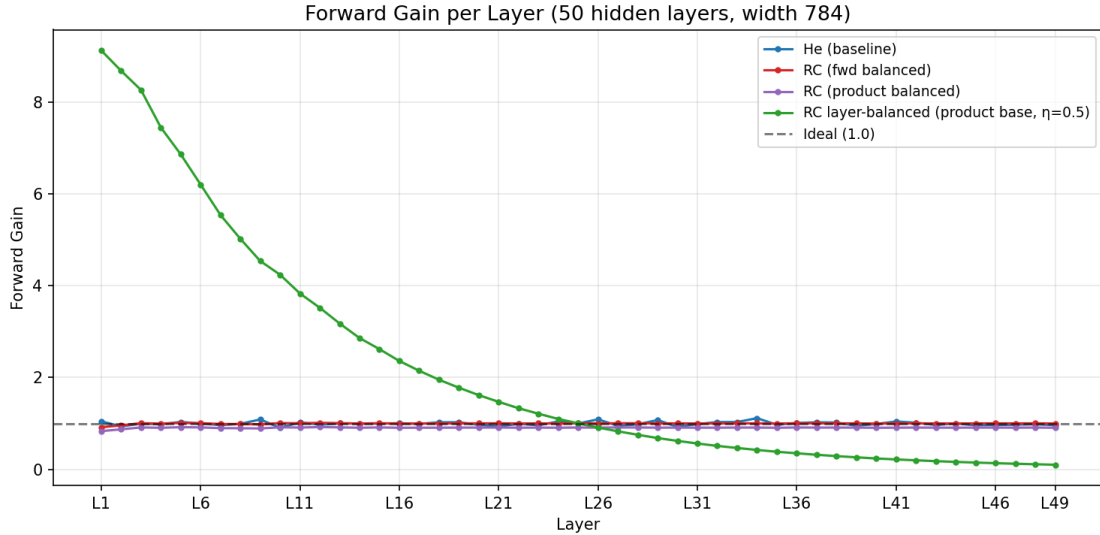


Figure 1: Forward gain per layer. Layer-balanced (green) sits between product-balanced and He.

5.4 Backward gain per layer

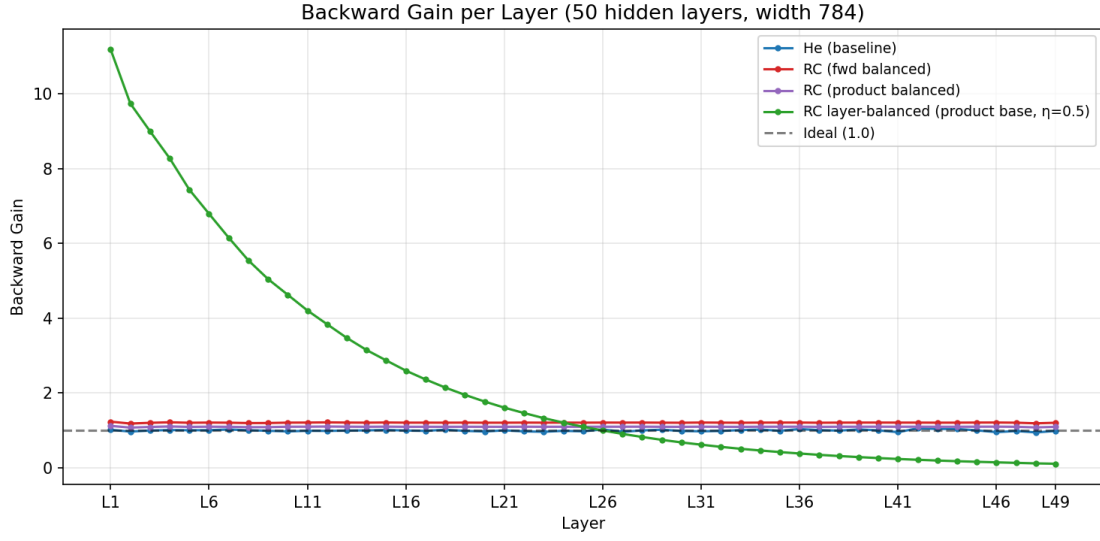


Figure 2: Backward gain per layer. Layer-balanced (green) has moderate backward gain ≈ 1.05 .

5.5 Gain product per layer

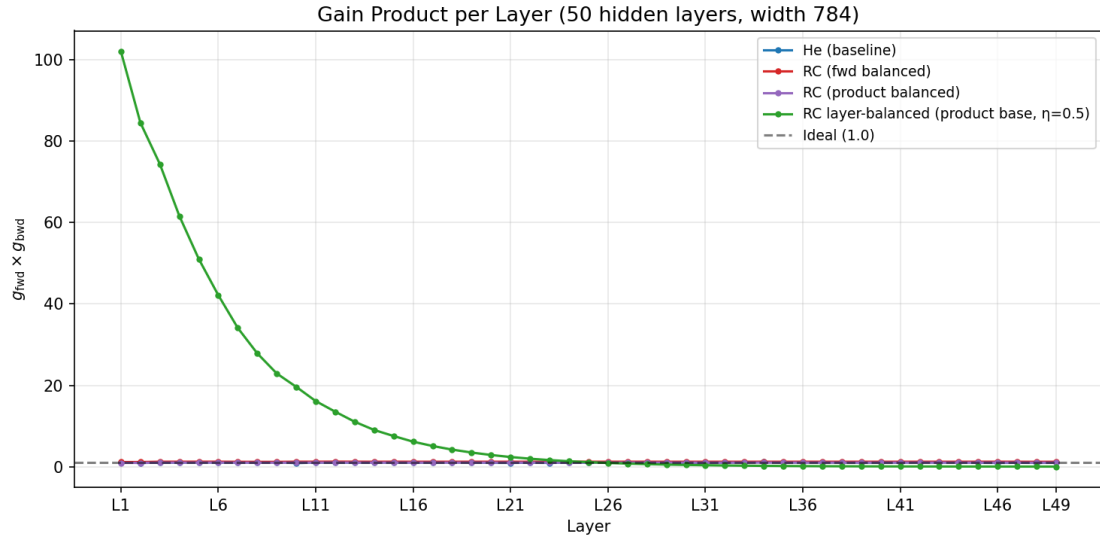


Figure 3: Gain product per layer. Product-balanced and layer-balanced both sit near 1.0.

5.6 Gradient row norms

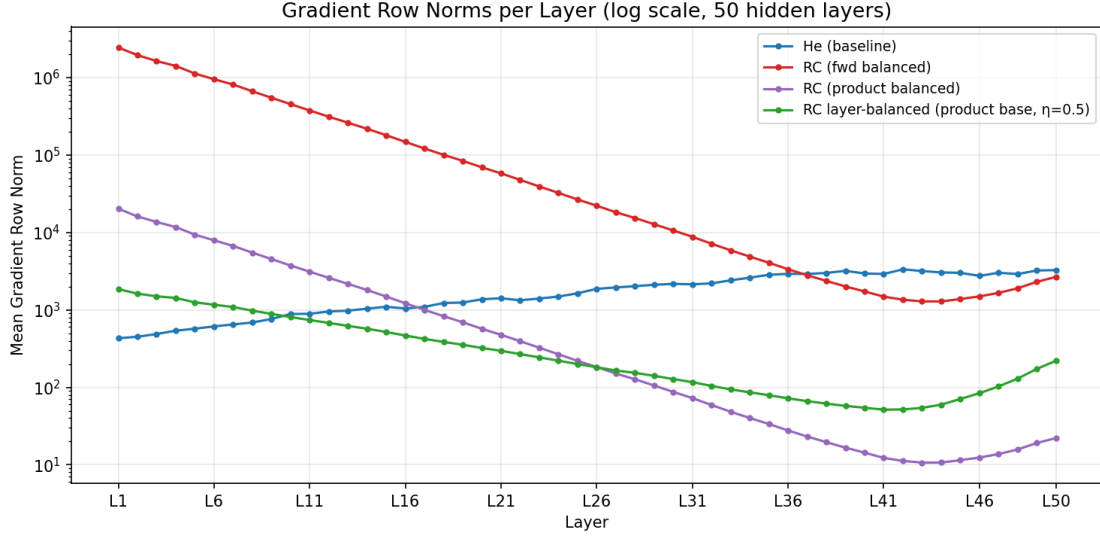


Figure 4: Gradient row norms (log scale). He is nearly flat. Layer-balanced (green) has moderate slope. Uniform RC variants (red, purple) have steep decay.

6 Geometry Preservation (k-NN Accuracy)

Table 2: k-NN accuracy vs. depth. Chance level = 0.10.

Initializer	5 L	10 L	15 L	20 L
He (baseline)	0.829	0.800	0.784	0.752
RC fwd-balanced	0.810	0.636	0.356	0.270
RC product-balanced	0.810	0.636	0.356	0.269
RC layer-bal ($\eta=0.5$)	0.810	0.636	0.356	0.270

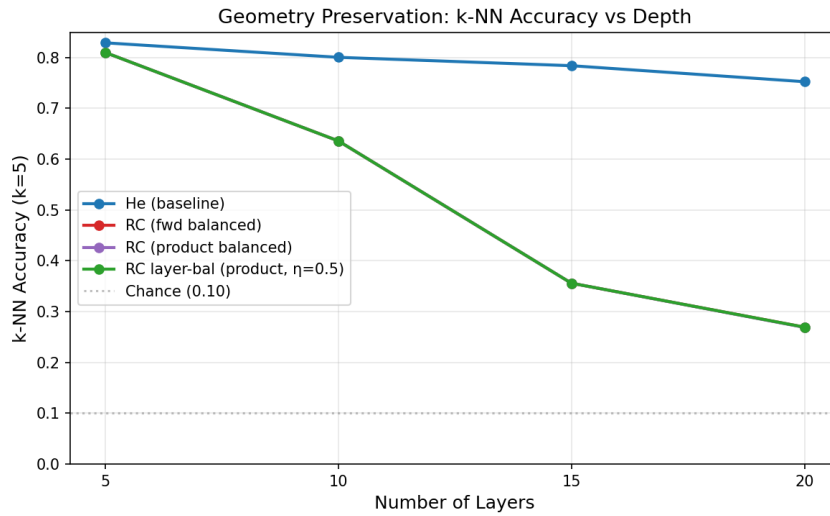


Figure 5: k-NN accuracy vs. depth. All three RC variants are identical.

Remark 6.1. All row-centered variants — uniform or layer-balanced, regardless of variance — produce identical k-NN curves. The constraint $\sum_j W_{ij} = 0$ makes the neuron blind to the mean activation. This DC-blindness is what destroys class structure, and it is completely independent of the variance scale s .

7 The Pareto Frontier

Theorem 7.1 (Pareto budget). *For the η -schedule (11):*

$$G(\eta) = r^{-(1-\eta)(L-1)} \quad (\text{gradient ratio}), \quad (13)$$

$$V(\eta) = r^{-\eta(L-1)} \quad (\text{variance ratio}). \quad (14)$$

Their product is independent of η :

$$\boxed{G(\eta) \cdot V(\eta) = r^{-(L-1)}}. \quad (15)$$

Proof. $G \cdot V = r^{-(1-\eta)(L-1)} \cdot r^{-\eta(L-1)} = r^{-[(1-\eta)+\eta](L-1)} = r^{-(L-1)}$. $\square$

Table 3: Pareto budget at various depths.

Depth L	Budget $r^{-(L-1)}$
10	$5.6\times$
20	$38\times$
50	$11,943\times$
100	1.7×10^8

Given a target gradient ratio T , the required η is:

$$\eta^*(L, T) = \max\left(0, 1 - \frac{\ln T}{(L-1) \cdot \ln(1/r)}\right). \quad (16)$$

8 Beyond Row Centering: What Are the Options?

The results above establish that row centering faces two fundamental barriers for deep networks:

1. **Gradient barrier:** the Pareto budget $r^{-(L-1)}$ grows exponentially with depth.
2. **Geometry barrier:** the DC-blindness ($\sum_j W_{ij} = 0$) destroys class structure independently of variance.

Both barriers stem from the same structural factor $r = \sqrt{(\pi-1)/\pi}$. To escape them, we must either change the weight constraint or change the nonlinearity.

8.1 The ReLU kernel contraction: an intrinsic limit

Even without row centering, every ReLU layer contracts angles. The arc-cosine kernel gives the expected cosine similarity after one layer:

$$\hat{K}(\alpha) = \frac{1}{\pi}(\sin \alpha + (\pi - \alpha) \cos \alpha), \quad (17)$$

with distortion $D(\alpha) = \hat{K}(\alpha) - \cos \alpha > 0$ for all $\alpha \in (0, \pi)$. This contraction is a property of ReLU itself — no initialization can eliminate it (Chapter 6 of the main report proves that no pointwise nonlinearity achieves $D(\alpha) = 0$ everywhere).

8.2 Orthogonal He: the current practical winner

Orthogonal initialization with He scaling ($W = \sqrt{2}Q$, where Q is from QR decomposition):

- $g_{\text{fwd}} \approx 1$, $g_{\text{bwd}} \approx 1$ (no gradient trap, no coupling).
- Same expected kernel as He — the systematic contraction $D(\alpha)$ is identical.
- k-NN accuracy: 0.662 at 20 L vs. He’s 0.639 — a $\sim 3.6\%$ improvement from reduced variance (better concentration around the expected kernel).
- No row centering $\Rightarrow$ no DC-blindness $\Rightarrow$ no geometry destruction via the 71° fixed-point mechanism.

The improvement of orthogonal over He is small because it only reduces the *variance* of the kernel estimate, not the *expected* distortion $D(\alpha)$.

8.3 Batch normalization: a dynamic alternative

The advisor’s observation that batch normalization (BN) works well for CNNs but not for fully connected networks is relevant here. BN performs at each layer:

$$\hat{a}_j = \frac{a_j - \mu_B}{\sigma_B},$$

where μ_B and σ_B are the batch mean and standard deviation. This is effectively a *data-dependent* centering and rescaling of activations — precisely what row centering tries to achieve via the *weights*.

The critical difference: BN centers the *actual activations* (adapting to the data at each layer), while row centering constrains the *weights* (a fixed structural constraint independent of data). BN can re-center without losing signal because it also rescales; row centering loses signal because the rescaling cannot recover the DC information that was structurally eliminated.

This suggests that the CNN/FC comparison experiments (with and without BN) may reveal whether BN’s dynamic centering avoids the Pareto frontier that static row centering cannot escape.

8.4 Summary of candidates

Table 4: Comparison of initialization approaches for deep ReLU networks.

Approach	Gradient stability	Geometry	Limitation
He	$g_{\text{fwd}} = g_{\text{bwd}} = 1$	Degrades (0.64 at 20 L)	ReLU contraction
Orthogonal He	$g_{\text{fwd}} = g_{\text{bwd}} = 1$	Slightly better (0.66)	Same expected kernel
RC product-balanced	Product = 1	Collapses (0.27 at 20 L)	Pareto + DC-blindness
RC layer-balanced	Ratio $36\times$ at $L=50$	Collapses (0.27 at 20 L)	Same barriers
BatchNorm (dynamic)	Adaptive	Unknown (to test)	Not an initializer

9 Conclusion

9.1 Results

1. The product-balanced variance $v^* = 2\sqrt{\pi/(\pi-1)}/d \approx 2.422/d$ achieves $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$ exactly.
2. Combined with per-layer scaling ($\eta = 0.5$), the gradient ratio at $L = 50$ improves from $1902\times$ to $36\times$.
3. The Pareto budget $G(\eta) \cdot V(\eta) = r^{-(L-1)}$ is a fundamental limit that grows exponentially with depth.

4. Geometry (k-NN accuracy) is identical for all RC variants — the variance is irrelevant.
5. Orthogonal He remains the practical winner for geometry + gradients, though it cannot eliminate the intrinsic ReLU kernel contraction.

9.2 Next steps

- CNN and FC experiments with and without BatchNorm, using He and candidate initializers, on Fashion-MNIST and CIFAR-10.
- Investigate whether BN’s dynamic centering avoids the Pareto frontier that static row centering cannot escape.